LLM-Driven Unified Framework with Grid Diffusion for Efficient Long Video Editing

Authors, *Staff, IEEE,*

***Abstract*—Extensive quantitative and qualitative experiments demonstrate the effectiveness of our approach, achieving superior editing consistency and efficiency compared to existing methods.**

***Index Terms*—Grid diffusion, Efficient long video editing**

I. Introduction

TEXT-guided video editing [1] aims to transform a source video into a new one in a generative manner, such that the resulting video semantically conforms to a target text prompt while simultaneously preserving some information from the source video as needed. In recent years, remarkable progress has been achieved in leveraging large-scale image datasets to train diffusion models [2], enabling the synthesis of high-fidelity images that are well aligned with textual descriptions. Building upon publicly available pre-trained Text-to-Image (T2I) diffusion models [3, 4], text-guided image editing has also advanced substantially [5].

In the video domain, a series of large-scale Text-to-Video (T2V) generative models have emerged, including close-sourced [6, 7, 8] and open-sourced [9, 10]. However, existing video diffusion models are generally vast in volume [11], especially, for long-form video generation and editing [1], current T2V models suffer from limited efficiency and flexibility due to memory and computational bottlenecks, although some methods claim that their models are extensible to handle longer video sequences [12, 13, 14]. Thus, for text-guided video editing task, many existing works instead adapt pre-trained T2I diffusion models to video data by additionally considering temporal-awareness information. These methods generally can be divided into two categories: zero-shot and one-shot video editing approaches. The former approaches [15, 16, 17] are particularly suitable for editing tasks that involve less structural changes (such as style transfer and color tone adjustment) in a training-free manner, while the latter [18, 19, 20] learn temporal-awareness information directly from the input video via per-video fine-tuning, enabling more complex edits (such as changes in object shape or position). Besides, no matter zero-shot or one-shot setting, different methods may be adept at doing editing of different spatial scales (local or global). Currently, several challenges in this area still exist. **Firstly**, there is still no unified framework that can flexibly handle diverse video editing tasks (zero-shot or one-shot, local or global) according to the specific circumstances. **Secondly**, many existing T2I-based video editing methods are limited to handling videos with at most several dozens of frames [16, 21, 22]. They often perform diffusion operations on an entire video bulk [18, 20], which severely restricts the length of videos that can be processed due to the computational limitation of the GPU. However, preliminary attempts in extending the length of the edited video inevitably compromises the editing performances [23, 24, 25] because of the inability of simultaneously keeping short-term and long-term temporal consistencies. **Thirdly**, the editing speeds of existing methods remain relatively slow. Although recent studies have explored various strategies (including token merging [21, 26, 27], caching and reuse [28, 29, 30, 31], and parallel computation [32, 33]) to accelerate diffusion model inference, the efficiency of long-video editing still falls short of the requirements for real-world applications.

To address the aforementioned challenges, we propose a Large Language Model (LLM)-driven **U**nified framework, which performs flexible grid manipulations based on a pre-trained T2I diffusion model to enable **E**fficient **L**ong-**V**ideo **E**diting across a wide range of editing tasks, termed as **UELVE**. Representative examples edited by UELVE are illustrated in Fig. 1, including object or background replacement, local attribute modification, global stylization, as well as motion preservation or editing. Large Language Models (LLMs) have recently been introduced to assist visual generation and editing tasks, however, existing methods primarily leverage LLMs to generate instructions and/or target prompts for constructing paired or triplet data to train instruction-based image/video editing models [34, 35, 36]. In contrast, we exploit the reasoning capability of a LLM to compare and analyze the source and target text prompts so as to identify the specific video editing targets and types. The LLM could serve as an agent to automatically make wise decisions between zero-shot and one-shot settings (involving whether per-video fine-tuning is required or not), as well as between local and global editing (involving whether spatial masks are needed or not). This design establishes the foundation for a unified, more precisely targeted, and cost-effective video editing framework.

Then, inspired by the grid trick (also known as the character sheet), a commonly used hack in the Stable Diffusion art community [3], we arrange the input video frames into a sequence of grids (e.g., each grid is with dimension 2×2, accommodating four frames) and perform sequential T2I diffusion by inputting one grid every time. Existing methods like [37] perform fully cross-frame interactions among all video frames, at the cost of substantial computational overhead. Some others adopt sparse causal attention [18], frame attention [20], or spatiotemporal self-attention [17] to model interactions between every frame and a single selected reference frame (the first, the immediately previous, or the middle frame), which however leads to insufficient modeling of local neighbor-

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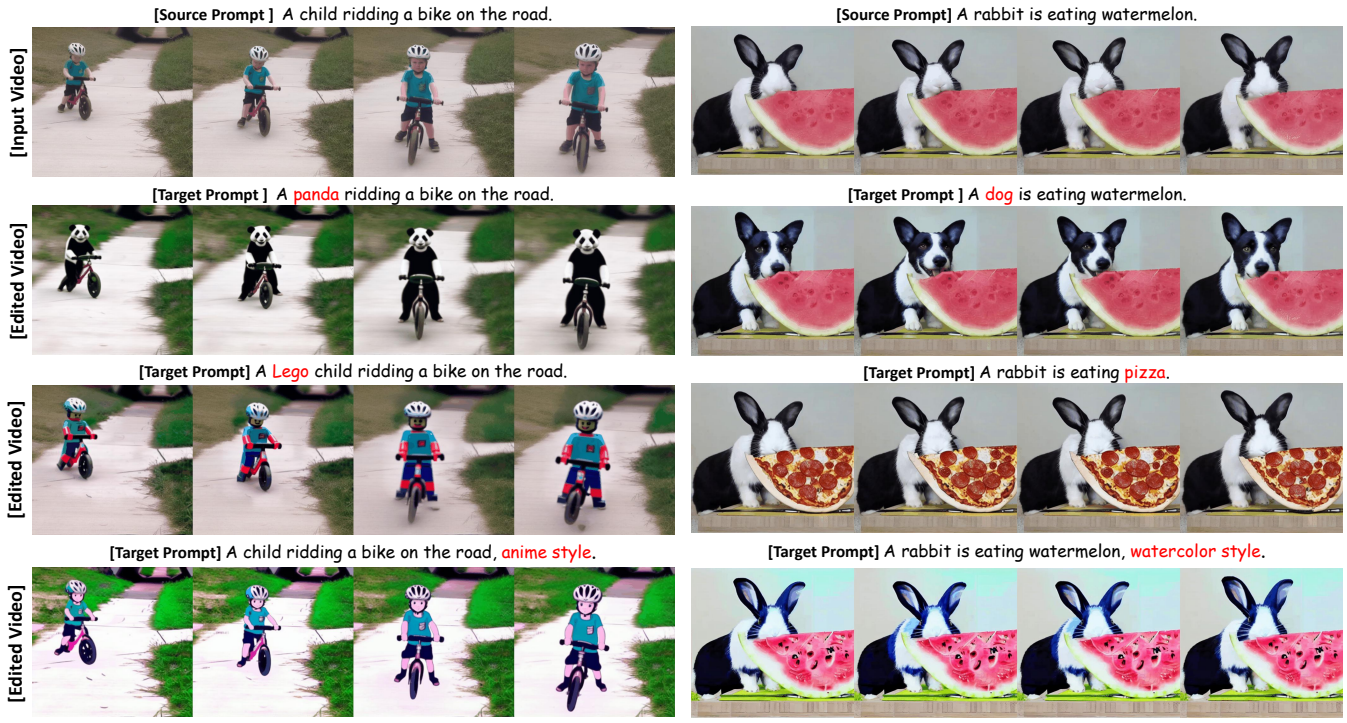

Fig. 1 **UELVE** is an efficient long-video editing method that leverages LLM to make decision on technical routes according to the specific editing targets and types, emphasizes both short-term and long-term temporal consistency by grid manipulations on a pre-trained Text-2-Image diffusion model, and enhance the efficiency by multi-GPU parallel computation with very little inter-GPU data communication. It can be used for object replacement, local attribute editing, global style transformation and so forth, while preserving some information as needed, such as background, semantic structure, and motion.

frame correlations. In contrast, UELVE leverages *grid self-attention* to explicitly capture neighbor-frame interactions within each grid, thereby ensuring short-term consistency. To further maintain long-term consistency, we introduce a simple yet effective *grid sliding manipulation*, which is applied to both the video inversion and denoising diffusion processes to enable inter-grid interactions. By sliding the adjacent grids back and forth between odd and even timestamps, the grid-based diffusion process effectively propagates neighbor-frame correlations throughout the entire video, ensuring temporal coherence and visual continuity. Based on such operations, UELVE decomposes a long video into manageable batches, enabling flexibly editing videos of arbitrary length in theory using only a single GPU in a memory-swapping manner.

Furthermore, UELVE also supports multi-GPU parallelization to accelerate the editing process, while requiring only a small amount of data communication (typically a few frames near the data division boundaries) between GPUs. This design not only reduces the overhead caused by large-scale inter-GPU communication (e.g., if data shuffling proposed in [16] is used on multiple GPUs), but also maintains temporal consistency across GPUs.

Extensive experiments demonstrate both quantitatively and qualitatively that UELVE achieves effective and efficient long-video editing across diverse editing tasks. To summarize, our main contributions are as follows:

1) We propose a novel unified diffusion framework for video editing, which leverages the reasoning capability of LLMs to make economical and practical decisions on technical routes according to the specific editing targets and types.
2) By organizing video frames into grids for batched processing, We introduce grid self-attention based on a pre-trained T2I diffusion model to enhance short-term temporal consistency across intra-grid neighboring frames. We also design a grid sliding manipulation to establish inter-grid connections during the diffusion process, maintaining long-term temporal consistency and thus enabling editing of long videos.
3) To enhance the efficiency, UELVE can be readily adapted to multi-GPU parallel computation to accelerate the editing process while maintaining cross-GPU temporal consistency, with only very little inter-GPU data communication.
4) UELVE is designed as a plug-in framework that can be seamlessly integrated with various existing image editing approaches, such as attention control [38] or ControlNet-based methods [39], to achieve diverse video editing effects. Extensive experiments, including both quantitative and qualitative evaluations, validate the effectiveness and efficiency of the proposed method for long-video editing.

II. Related Work

This section provides a review of the literature, focusing primarily on recent advancements in text-driven video generation and text-driven video editing.

A. Text-guided Video Generation and Editing

Diffusion models have demonstrated exceptional capabilities in video generation mainly by integrating modulated spatial-temporal modules [40, 41, 42, 43, 44, 45, 46, 47]. For instance, a video latent diffusion model (Video LDM) is first pre-trained on large-scale image datasets, which is afterward transformed to a video generator by introducing a temporal dimension into the latent space of DM. It entails training the temporal layers on image sequences, i.e., videos, while fixing the spatial layers. To reduce the text-to-video training cost, make-A-Video [42] decomposes the full temporal U-Net and attention tensors and approximate them in space and time. They also design a spatial temporal pipeline to generate the video frames with a video decoder and an interpolation model. However, this method still requires massive training on a large dataset, limiting its generalisation and scalability. Unlike instruction-based image editing methods that require the use of LLM to generate a large number of instructions and target prompts, construct a large-scale dataset, and then conduct large-scale training, our UELVE just needs LLM to analyze the changes between source and target prompts to make appropriate choices for the techniques used, which is more cost-effective and pragmatical.

Text-to-image (T2I) editing refers to the process of generating or modifying images based on textual descriptions. This involves using natural language input to instruct a computer system to create, alter, or manipulate visual content, and thus has garnered tremendous attention due to its significance in various applications. Inspired by the CLIP [48], which offers strong semantically joint text-image representation trained over millions of text-image pairs, recent works can revolutionize the landscape by adapting the text-image alignment without requiring any manual input and produce realistic manipulations using texts only.

In a nutshell, text-to-video (T2V) editing approaches can be categories into two streams: 1) fine-tune pre-trained text-to-image diffusion models with additional training on target prompt; and 2) zero-shot video editing without additional training on target prompt. Numerous approaches in T2V are developed to enhance the edit in both global and local details [49, 50]. For instance, [49] uses an architecture via a factorized space-time U-Net which is compatible with joint image-video training. As a bootstrap to Dreamix [50], [49] is extended with a mixed fine-tuning approach in Dreamix [50]. Several recent methods are dedicated to improve the local edit by using spatial temporal attention interactions between textual tokens and latent vectors [18, 20, 51, 52]. For instance, akin to [20], [52] invents "sparse-causal blending" to inhibit the background inconsistency, making use of the null-text inversion [53].

Due to the optimization requirements of previous methods, recent studies emphasize zero-shot training-free approaches for practical applicability. By utilising the predictions of the original images as a proxy at each denoising step, Pix2Video [54] employs the sparse-causal attention for temporal consistency along with the latent guidance. FateZero [17] advocates the attention features during inversion for spatial-temporal preservation and blending where it has been demonstrated that attention features contingent on inversion are superb in preserving motion and structure compared to that of during sampling. A follow-up effort of Text2Video-Zero [55] synthesizes and edits videos with a bulk of cross-frame attention, initial frame integration, and background smoothing. Rerender-A-Video [56] employs hierarchical cross-frame constraints for temporal consistency, while TokenFlow [15] focuses on feature-level smoothing to reduce the effects of flickering. Nevertheless, FateZero [17] as well as Pix2Video [54], require source prompts due to the employment of the Prompt-to-Prompt [38] editing technique. This paradigm necessitates prompts characterised of the source prompt, and resulting in limited editing diversity. Moreover, FateZero [17] and Pix2Video [54] are bounded to shorter clips due to memory constraints. Unfortunately, the reliance on optical flow has lent Text2Video-Zero [55] and Rerender-A-Video [56] short in portrait consistency over longer videos [44]. TokenFlow [15] is struggling with structural editing, which is caused by the low inversion quality, and feature-level smoothing with blurring edges. As opposed to aforementioned methods that require time-intensive operations and thus slow down inference speed, we tackle the long video rendering by using fast video inversion based on random shuffling for the applicability to longer videos.

B. LLM-Driven Video Editing

A concurrent work, known as RAVE, ...

C. Grid diffusion

Grid diffusion [57]

D. Diffusion model acceleration

III. Methodology

A. Overview

Let $\mathcal{V}_K = \{\mathcal{I}^1, \ldots, \mathcal{I}^K\}$ be an input video containing K frames, which is paired with a source prompt $\mathcal{P}$, and $\mathcal{I}^k \in \mathbb{R}^{W \times H \times C}$ denote the k-th frame in $\mathcal{V}_K$ where W, H and C specify the width, height and channel, respectively. Given a target prompt $\mathcal{P}^*$, the video editing task is to translate the source video $\mathcal{V}_K$ into a modified version $\mathcal{V}_K^*$, which is semantically aligned with $\mathcal{P}^*$ while preserving some basic information of $\mathcal{V}_K$ as needed. In this work, based on a LLM-driven unified framework with grid manipulations on a pre-trained T2I diffusion model, we aim for efficient long video editing covering diverse specific editing targets and types.

Our proposed UELVE framework is shown in Fig. 2. We first arrange the frames of input video into grids, encode the grids via Variational AutoEncoder (VAE) [58] conforming to the paradigm of Latent Diffusion Models [3], and then run DDIM inversion [59] via T time steps of iterations, conditioned on the source prompt, to acquire the inverted noisy

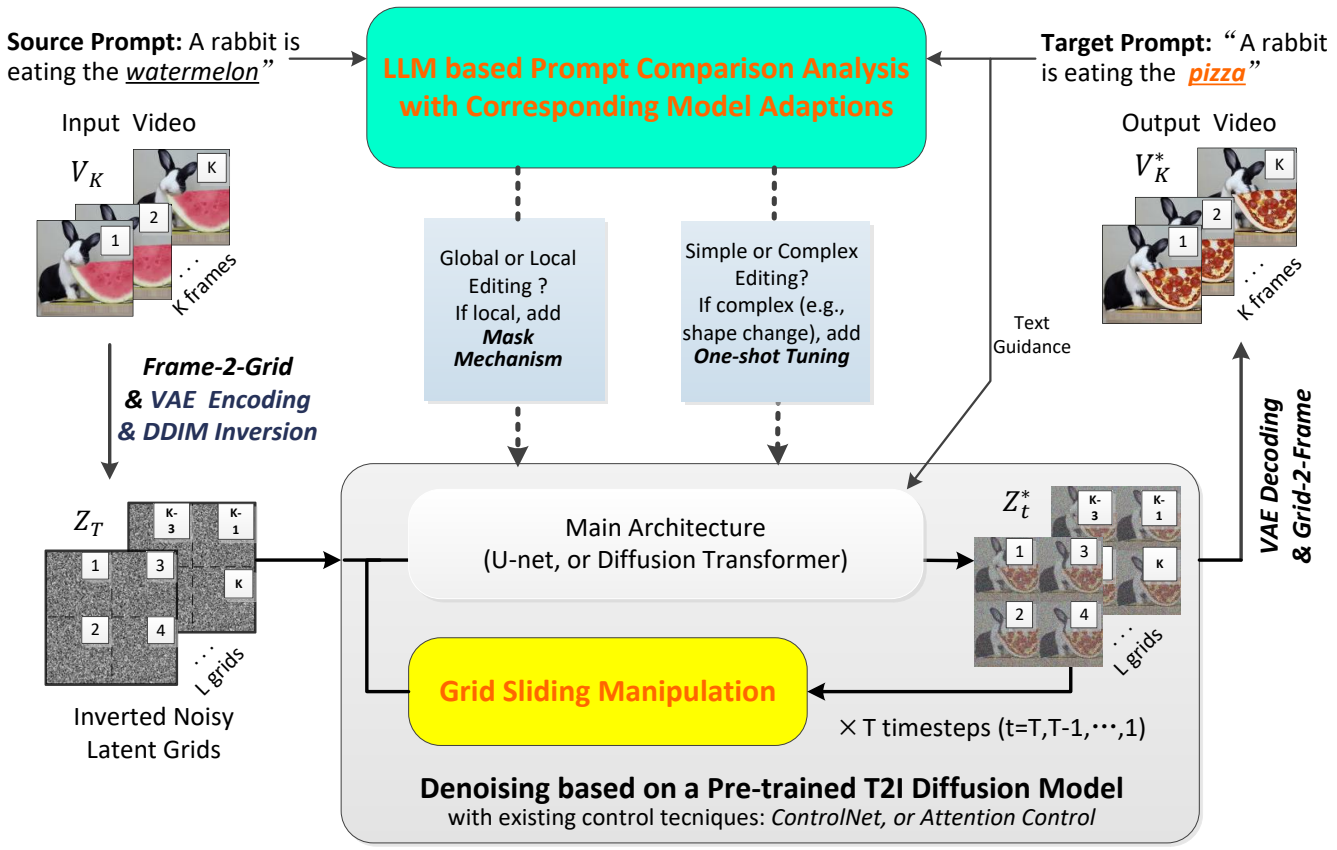

Fig. 2 **Framework of UELVE**. We first arrange the frames of input video $\mathcal{V}_K$ into grids (Frame-2-Grid), encode them into the latent space (VAE Encoding) and then invert the original latents to Z_T (DDIM Inversion). Then we denoise the grid latents Z_T to Z_0^* via T time steps of iterations on a pretrained Text-to-Image diffusion model, decode Z_0^* (VAE Decoding) and rearrange the grid frames into an edited video $\mathcal{V}_K^*$ as output (Grid-2-Frame). For the denoising process, by comparing the source and target prompts, we leverage a LLM to reason out specific editing targets and types and thus make decisions on corresponding model adaptions, unifying diverse editing tasks into a same framework. Beyond the grid self attention for within-grid interactions, we design a grid sliding manipulation to enhance cross-grid interactions to ensure the editing quality of long videos. Benefiting from the flexible grid manipulations, a multi-GPU data parallel computing strategy is also considered to remarkably improve the computation efficiency.

latent grids. Next, starting from the inverted noisy latents, we run grid diffusion denoising conditioned on the target prompt for T steps of iterations as well. The latent grids obtained at the last time step go through VAE decoding and then are rearranged into pixel frames chronologically, forming the edited video. It's worth noting that, for the diffusion process based on the pre-trained T2I model, beyond the traditional text prompt guidance, we leverage a LLM to compare the source and target text prompts, and analyze the specific editing targets and types, thus reason and make decisions on zero-shot or one-shot setting, local or global editing depending on whether fine-tuning is needed or not, whether mask is needed or not, respectively. Our framework could do corresponding model adaptions according to specific circumstances, such as integrating fine-tuning techniques for one-shot setting, or introducing masking mechanisms for local editing (Section III-C). Besides, the grid self-attention make it possible to facilitate the neighbor-frame interactions within each grid to maintain the short-term consistency, on the other hand, to ensure the long-term consistency, we design a simple yet effective grid sliding manipulation to achieve inter-grid interactions (Section III-D). Moreover, our grid-based diffusion models are easy to be deployed on multiple GPUs to enable flexible data parallel computing, and by the grid sliding near the data division boundaries, only very little inter-GPU data communication is needed to maintain the cross-GPU consistency, which accelerates the diffusion process significantly (Section III-E).

B. Preliminaries

a) Latent Diffusion Models (LDMs): LDMs [3] typically run the diffusion models (such as denoising diffusion probabilistic models (DDPM) [2]) in a lower-dimensional latent space produced by a pre-trained autoencoder, Variational AutoEncoder (VAE) [58], to significantly reduce the resources required for pixel-level calculation. In specific, the encoder $\mathcal{E}$ of VAE is used for compressing images (or video frames)

x into latent representations $z = \mathcal{E}(x)$, and in the end of all the diffusion operations the decoder $\mathcal{D}$ reverts the latent (e.g., z^*) to pixel space, i.e., $\mathcal{D}(z^*) = x^*$. In the latent space, given the data $z_0 \sim q(z_0)$, the forward diffusion process is subject to a Markov Chain, which progressively corrupts z_0 to get $z_1, ..., z_T$. The Markov transition $q(z_t|z_{t-1})$ is defined as a Gaussian distribution with a variance schedule $\beta_t \in (0, 1),\ t = 1, ..., T$, that is,

$$q(z_t|z_{t-1}) = \mathcal{N}(z_t; \sqrt{1-\beta_t}z_{t-1}, \beta_t I) \tag{1}$$

According to Markov property, the conditional probabilities $q(z_t|z_0)$ can be acquired:

$$q(z_t|z_0) = \mathcal{N}\Big(z_t; \sqrt{\overline{\alpha}_t}z_0, (1-\overline{\alpha}_t)I\Big) \tag{2}$$

where $\alpha_t = 1 - \beta_t$, $\overline{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. While in the backward denoising process, the model is to turn the noise into the original data distribution gradually. This process can be described as follows:

$$\begin{aligned} p_\theta(z_{t-1}|z_t) &= \mathcal{N}\Big(\mu_\theta(z_t, t), \Sigma_\theta(z_t, t)\Big) \\ &= \mathcal{N}\bigg(z_{t-1}; \frac{1}{\sqrt{\alpha_t}}\Big(z_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(z_t, t)\Big), \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t}\beta_t I\bigg) \end{aligned} \tag{3}$$

where $\epsilon \in \mathcal{N}(0, I)$ and θ denote the parameters of the denoising neural network. In training stage, the objective is to maximize the likelihood of observed data $p_\theta(z_0) = \int p_\theta(z_{0:T})dz_{0:T}$ by maximizing its evidence lower bound (ELBO). The network is trained to realize the parameterized $p_\theta(z_{t-1}|z_t)$ to approximate the theoretical denoising process $q(z_{t-1}|z_t, z_0)$ which can be derived theoretically via Bayes' rules. Through simplification, the final objective is to train a network ϵ_θ to predict an artificial noise:

$$\min_\theta E_{z_0, \epsilon\sim\mathcal{N}(0,I), t\sim Uniform(1,T)}||\epsilon - \epsilon_\theta(z_t, t, \mathcal{C})||_2^2 \tag{4}$$

where z_t is a noised sample and the noise is added to the clean data z_0 according to timestamp t, and $\mathcal{C} = \Phi(\mathcal{P})$ is the embedding of the text prompt condition which can be injected into $\epsilon_\theta(\cdot)$. At inference, given a random noisy vector z_T, the noise can be progressively removed by sequentially predicting $\epsilon_\theta(z_t, t, \mathcal{C})$ using the trained network for T steps. That is, by iteratively sampling z_{t-1}, the denoising process can generate a new sample z_0^* finally.

b) Denoising Diffusion Implicit Models (DDIM): DDIMs [59] are variants of DDPM [2], which can accelerate the denoising (sampling) process, belonging to a more efficient class of iterative implicit probabilistic models with the same training procedure and objective as DDPMs. The DDIM denoising is expressed as follows:

$$z_{t-1} = \sqrt{\frac{\alpha_{t-1}}{\alpha_t}}z_t + \sqrt{\alpha_{t-1}}\left(\sqrt{\frac{1}{\alpha_{t-1}} - 1} - \sqrt{\frac{1}{\alpha_t} - 1}\right)\epsilon_\theta(z_t, t, \mathcal{C}) \tag{5}$$

In contrast to DDPMs, the denoising generative process of DDIMs is deterministic, which is a desirable property in the editing for real image/video. A simple inversion can be derived based on the ordinary differential equations (ODEs) limit analysis of the diffusion process, a.k.a. DDIM inversion, which can map a clean latent z_0 (encoded from real pixel data via VAE) to a noised latent $\hat{z}_T$ in limited time steps:

$$z_{t+1} = \sqrt{\frac{\alpha_{t+1}}{\alpha_t}}z_t + \sqrt{\alpha_{t+1}}\left(\sqrt{\frac{1}{\alpha_{t+1}} - 1} - \sqrt{\frac{1}{\alpha_t} - 1}\right)\epsilon_\theta(z_t, t, \mathcal{C}) \tag{6}$$

C. LLM-driven Unified Diffusion Framework

Existing video editing methods [15, 16, 17, 18, 20] can typically divide into zero-shot and one-shot methods. Zero-shot methods [15, 16, 17] struggle to maintain temporal consistency through a training-free approach, limiting them to tasks with minimal structural changes, such as style transfer and color adjustment. On the other hand, one-shot methods [18, 20] can handle more complex editing tasks like object replacement, but must fine-tuning the pre-trained model on each input video. In terms of the editing type, video editing tasks can be categorized into local and global editing. Local editing need masks to separate foreground from background to prevent unintended modifications in background region, whereas global editing does not. These categories of approaches have their respective strengths and weaknesses. Existing video editing approaches often excel in only one or a limited subset of the four categories mentioned above. There still don't exist a unified framework that can combine the advantages of them to handle various types of video editing tasks.

Recently, LLM have been used to assist visual generation and editing tasks, they possess powerful semantic understanding capabilities and reasoning ability. we leverage the reasoning ability and semantic understanding capabilities of LLM to analyze the text prompts, determine the target and specific types of video editing, and then make appropriate decisions between zero-shot and one-shot operations, local and global editing, forming the premise for a unified video editing framework. Considering the feasibility, stability, practicality, and cost, we select the DeepSeek as our LLM via its official API. The pipeline for LLM to make decision and guide the video editing as follows. Firstly, foundational knowledge, including the basic theory of the diffusion model, the video editing pipeline, and the mechanism of the attention control method, is provided to the LLM as a fixed knowledge library. Furthermore, the Editing Type Classification Standard that we summarize from existing video editing methods (AS illustrated in Fig. 16) also is given to the LLM as the classification guidance. Then, LLM performs a semantic analysis to current task's text prompts (comprising both the source and target prompts). It classifies the editing task described by the text prompts based on the Editing Type Classification Standard. Finally, grounded in the fixed knowledge library, classification result, and semantic analysis, the LLM automatically makes an informed decision, which then directs the video editing model to perform the specific video editing task. Specifically, the decision making operation focus on two core point, the extent of structure changes which determine using zero-shot or one-shot and the editing type is local or global which determine using mask mechanism or not. Furthermore, existing video

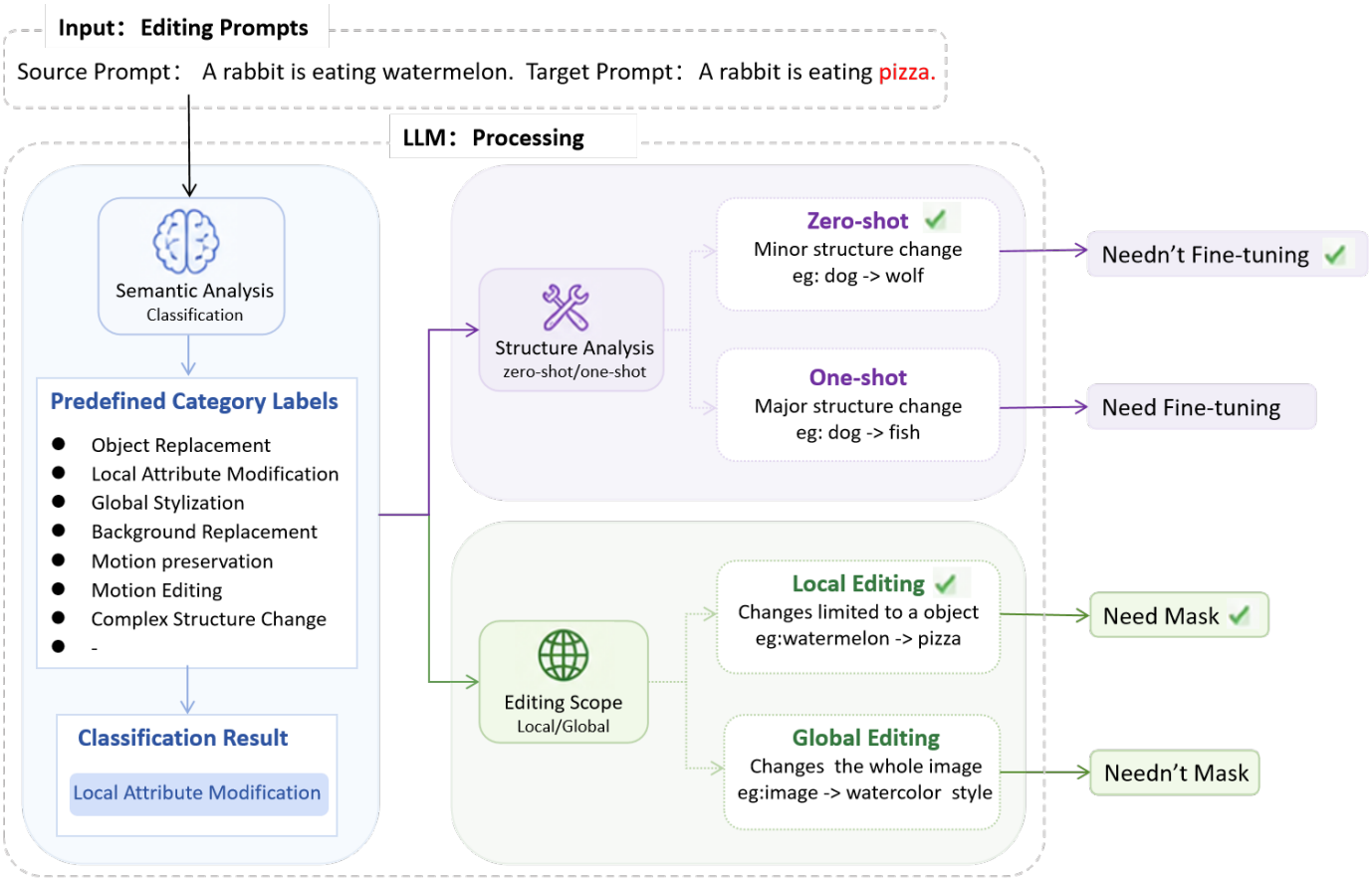

Fig. 3 **The illustration of LLM Processing User-Provided Prompts**. We input text prompts into the LLM. After performing semantic analysis, the LLM classifies the editing operations described by the text prompts. On this basis, it infers the editing scope and the degree of structural change of the edited objects, and determines whether masking and fine-tuning are required.

editing methods [17, 18, 20] require users to manually construct parameters configuration files based on the text prompts that not only consumes time and energy, but is also not user-friendly for non-professional users. However, our UELVE can intelligently recommend values for these parameters by LLM, as shown in Fig 17.

D. Grid Manipulations

It's worth noting that, [1].

Although the consistency within grids is attained by grid self attention and the temporal attention, the diffusion trajectories of different latent grids diverge over the time steps, leading to disparity across grids. Especially, when confronting with long videos involving variations of scenes or objects, it becomes challenging to maintain the long-term consistency while allowing for variations, though with the same text prompt. To this end, we propose a grid layout alternation strategy to be used in both the inversion and the denoising processes as shown in Fig.2. To be specific, we present two basic grid layouts, namely, patterns of arranging the grid latents, and use them

[1]whether it is also necessary to apply the grid sliding and mask manipulations in to the inversion process, it depends on, e.g., the grid sliding is applied during both inversion and denoising when using the attention control method, but only during denoising when using ControlNet method.

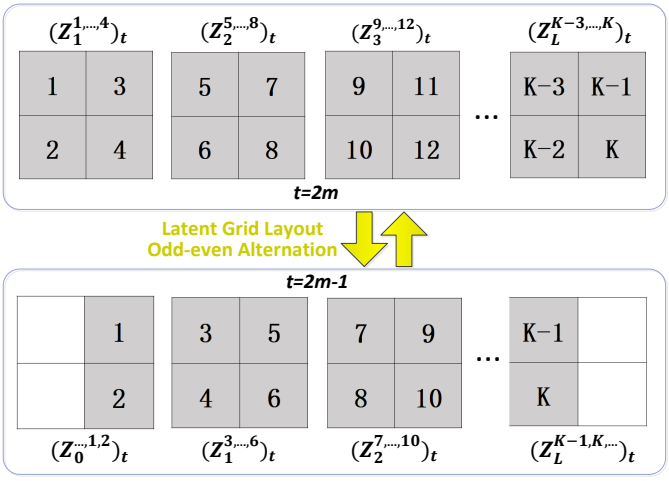

Fig. 4 **The illustration of grid layout alternation** between odd-even time steps of video *inversion* ($m = 0, 1, ..., T/2$) or *denoising* ($m = T/2, T/2-1, ..., 0$) diffusion process.

alternately in diffusion process. The first is to just stay the original grid layout chronologically as follows:

$$\text{GridLayout}_1 = \left\{ \mathcal{Z}_1^{1,\ldots,N}, \ldots, \mathcal{Z}_L^{P,\ldots,K} \right\} \tag{7}$$

while the other layout is as follows:

$$\text{GridLayout}_2 = \left\{ \mathcal{Z}_0^{1,\dots,n}, \mathcal{Z}_1^{n+1,\dots,N+n}, \dots, \mathcal{Z}_L^{P+n,\dots,K} \right\} \tag{8}$$

where n denotes the number of frames arranged in a column of the normal grid of size $n \times l$. Compared to GridLayout_1, it is to take the first grid column out of $\mathcal{Z}_1^{1,\dots,N}$ separately as an additional grid $\mathcal{Z}_0^{1,\dots,n}$, and all the other grids *slide* forward a column in order, with the last grid $\mathcal{Z}_L^{P+n,\dots,K}$ lacking a column. To complement $\mathcal{Z}_0^{1,\dots,n}$ and $\mathcal{Z}_L^{P+n,\dots,K}$ to form a complete grid of same size $n \times l$ so that they can go through all the layers of U-net like other grids, we pad the absent grid columns with repeated frames from themselves. Then the alternation operation is defined as:

$$\begin{aligned} &Alternate(\text{GridLayout}_1) = \text{GridLayout}_2, \\ &Alternate(\text{GridLayout}_2) = \text{GridLayout}_1 \end{aligned} \tag{9}$$

When acquiring GridLayout_1 from GridLayout_2 by $Alternate(\cdot)$, the padding can be deleted for subsequent computing.

Then we can use the operator Eq.9 to rearrange the grid layout alternately in each diffusion step, so that the obtained layouts after the alternation operation present different forms in the odd and even time steps, that is,

$$\begin{cases} (\mathcal{Z}_1^{1,\dots,N})_t, \dots, (\mathcal{Z}_L^{P,\dots,K})_t, & t = 2m \\ (\mathcal{Z}_0^{1,\dots,n})_t, (\mathcal{Z}_1^{n+1,\dots,N+n})_t, \dots, (\mathcal{Z}_L^{P+n,\dots,K})_t, & t = 2m-1 \end{cases} \tag{10}$$

In summary, the grid layout of the whole set of latents will alternate repeatedly between the odd and even time steps of the *inversion* ($m = 0, 1, ..., T/2$) or *denoising* ($m = T/2, T/2-1, ..., 0$) diffusion process, as shown in Fig.5, we takes the grid of size 2×2 for example. Through such a latent grid layout alternation strategy, the layout rearrangement indirectly achieves inter-grid interactions, encouraging the connections across multiple grids and thus maintaining long-term consistency throughout all the frames.

E. Multi-GPU Data Parallel Computing

The Grid Layout Alternation Strategy mentioned above enhances interactions among different grids, thereby effectively maintaining long-term consistency in videos. However, when processing longer videos, the computational efficiency of a single GPU become bottlenecks. To address this, we further extend the Grid Layout Alternation strategy to a multi-GPU parallel computing environment. Assume we use G GPUs to process a video containing K frames. The entire frame sequence must be uniformly divided into G segments, with each GPU assigned one segment containing a contiguous subsequence of $M = K/G$ frames and the grid shape is nxl.

If each GPU independently performs video editing task on its assigned frame subsequence, the lack of cross-GPU interaction will prevent consistency between GPUs and result in noticeable visual discontinuities between segments in the edited video. Therefore, we introduce a cross-GPU boundary frame interaction mechanism driven by the Grid Layout Alternation Strategy at each diffusion timestep. More precisely,

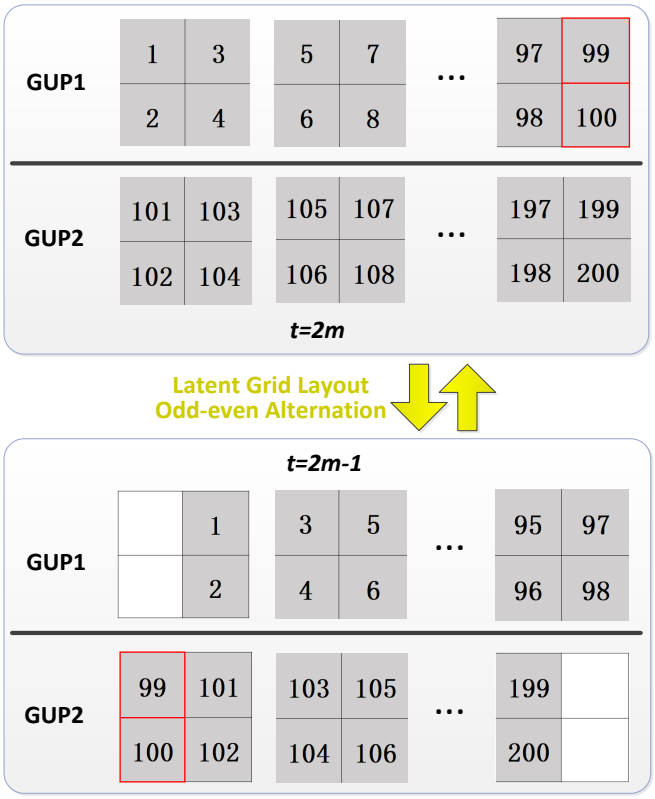

Fig. 5 **The illustration of Interaction between Two GPUs**

when switching from an even timestep (Layout_1) to an odd timestep (Layout_2), the last n frames of GPU_g are transferred to GPU_{g+1}, where $g = 1, 2, \dots, G-1$. Conversely, when switching back from an odd timestep to an even timestep, these frames are returned to their original GPU. This periodic exchange ensures that frames near GPU boundaries continuously participate in the attention computations of adjacent GPUs, thereby preserving visual and temporal continuity throughout the entire video.

To summarize, we further extend the Grid Layout Alternate Strategy to a multi-GPU parallel computing environment. This approach not only ensures consistency across GPUs but also significantly enhances computational efficiency. With G GPUs, the computational efficiency performance for video editing increases by nearly G times, which outperforms most other acceleration methods.

In addition, most existing methods lack an effective GPU interaction mechanism, which hinders the application of multi-GPU data parallelism. Although some acceleration libraries offer model-parallel training, they impose strict constraints on the model architecture and deliver far inferior performance compared to data-parallel methods. Notably, our UELVE method only need to exchange minimal data (typically, a couple of frames near the Data break point), whereas other methods (e.g., using shuffling in RAVE [16])) need large data communication (More details reference VI-A3). Consequently, the acceleration achieved by RAVE [16] is significantly less than that of our UELVE. As evidenced by Table I, UELVE stands out as the only method suitable for multi-GPU data-

parallel video editing, while all other baselines are unsuitable.

Table I The suitability of baseline models for using Multi-GPU data parallel method is examined from the three stages of fine-tuning, inversion, and denoising. Here, ✓ indicates suitable, ◯ indicates that can be used but with Suboptimal result, and × indicates unsuitable. In addition, —indicates that the method doesn't have the stage.

Method	Tune	Inversion	Denoising
Tune-A-Video [18]	◯	×	×
Video-P2P [20]	◯	×	×
FateZero [17]	◯	×	×
RAVE [16]	—	◯	◯
Control-A-Video [60]	—	—	×
ControlVideo [37]	◯	×	×
TokenFlow [15]	—	×	×
AdaFlow [23]	—	×	×
GLFM [61]	—	×	×
UELVE-Attention(zero-shot)	—	✓	✓
UELVE-Attention(one-shot)	✓	✓	✓
UELVE-ControlNet	—	✓	✓

IV. Experiments

A. Experiment Details

1) Dataset: Currently, research on video editing is gradually maturing, but there is no widely popular video dataset adopted by most video editing methods. Different video editing methods select dataset suited to their specific characteristics, lacking a unified standard dataset. Thus, we organized a comprehensive video dataset named **UELVE-LD** including 150 videos. This dataset contains videos of varying lengths, systematically composed of the following three types of materials:

Benchmark Video Datasets: We selected representative video clips from publicly available datasets that are widely used in existing video editing research, including LongV-EVAL [23], VE-Bench [62], RAVE [16], Tune-A-Video [18], and Video-P2P [20], consisting of 50 videos. The textual descriptions were given by the corresponding datasets.

Real-World Footage: This subset consists of professionally shot videos provided by the authors and documentary clips sourced from the Internet, including 70 videos. The corresponding textual descriptions were generated by a large language model (LLM) and manually verified by researchers.

AI-Generated Content: Synthetic videos extracted from video generation models, covering a variety of visual styles, including 30 videos. The corresponding textual descriptions were given by the video generation models and manually verified by researchers.

2) Implementation Details: We adopt the Stable Diffusion v1-5 from the official Hugging Face repository as the pre-trained model. To ensure a fair comparison, all baseline methods use the pretrained models specified in their corresponding papers. In all comparisons (including quantitative comparison, qualitative comparison, and user study), we apply depth-conditioned ControlNet for all ControlNet-based method, and set grid shape with 2×2 for our method and RAVE [16].

During the training (fine-tuning) phase, we employ the Adam optimizer with a learning rate of 3e-5 and train for 300 timesteps. The sampling phase consists of two parts: grid inversion and grid denoising, both of which use a fixed classifier-free guidance scale of 12.5 and run for 50 timesteps. The primary grid shapes used include: 2×2, 2×4, and 3×3. It is particularly important to note that the grid shape during the training phase should match that during the denoising phase. To ensure fairness, neither our model nor the baseline models use any positive or negative prompts. All experiments are conducted on a server equipped with two 24GB GPUs.

B. Comparisons With The State-of-the-art Methods

1) Qualitative Comparisons: we compare our UELVE with nine state-of-the-art video editing methods. Including Tune-A-Video [18], Video-P2P [20] (Due to the GPU memory limitations, we use its fast mode.), FateZero [17], RAVE [16], Control-A-Video [60], ControlVideo [37], TokenFlow [15], AdaFlow [23], and GLFM [61]. we also try the long video editing method LoveCon [63], but it can only handle video with less than 30 frames. Therefore, we do not include it as a baseline method.

As shown in Fig. 6, we compare the editing results of our UELVE method with baseline methods on evaluation videos. The performance of our UELVE method significantly outperforms the baseline methods when using the same base model. Both the sparse causal attention mechanism of Tune-A-Video [18] and the frame attention mechanism of Video-P2P [20] are coarse-grained self-attention mechanisms. They all fail to model the car's motion information, resulting in inadequate representation of the visual effect of the car moving from far to near. ControlNet-based methods lack an effective masking mechanism, resulting in backgrounds in an overly consistent yet distorted uniform style. Specifically, RAVE [16] randomly rearranges frame sequences, exacerbates alterations in background regions. ControlVideo [37] suffers from blurred backgrounds due to the interference among multiple control conditions. Control-A-Video [60]'s reward feedback learning algorithm optimizes motion consistency, but intensifies changes in background areas. TokenFlow [15] and AdaFlow [23] rely on the token correspondence map to back-propagate features from keyframes to all frames, but the incorrect positional mappings often cause unintended background alteration in local editing tasks, e.g., the color of the trees. GLFM [61] compresses visual information (tokens) into compact basis vectors, which blend foreground and background features, causing the editing instructions to modify the entire scene. (the trees in background area are severely altered).

2) Quantitative Comparisons: **Effectiveness: Performance Comparisons**

We randomly select a large amount of videos from our UELVE-LD dataset and perform video editing tasks according to the corresponding text prompts in dataset. We evaluate the UELVE model and baseline models using various metrics. Masked-PSNR (Masked Peak Signal-to-Noise Ratio) evaluates the background preservation capability of methods. SSIM (Structural Similarity Index Measure) measures the structure

Fig. 6 Qualitative comparison with baseline methods. The video material come from our UELVE-LD dataset in AI-generated type. The baseline method cross two key technical paradigms: zero-shot/one-shot methods and ControlNet-based/Attention Control-based/Feature Propagation-based methods.

similarity between the frame image of source video and edited video. Text Alignment assesses the alignment between video frames and their text description using the CLIP model. Tem-Con(Temporal Consistency) assesses the continuity of features between adjacent video frames. (the specific functions and calculation methods of each metric are detailed in VI-A1). The average values are taken as the final evaluation results.

As shown in Table II, Video-P2P [20] is a one-shot method, while FateZero [17] is a zero-shot method. After fine-tuning, the model becomes more proficient in adjusting the contour and structure of edited objects according to the target prompt. Therefore, compared to FateZero [17], Video-P2P [20] achieves lower Masked-PSNR and SSIM metrics but exhibits higher Text Alignment metric. ControlNet-based methods utilize control images (e.g., depth, Canny edge maps) rather than mask to constrain the edited video structure, resulting in higher performance on the Tem-Con metric but poorer performance on Masked-PSNR and SSIM metrics. Interestingly, because lacking the mask mechanism, ControlNet-based models can

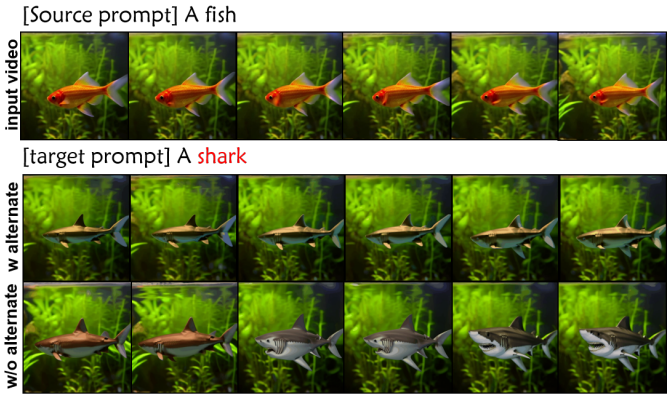

Fig. 7 The ablation experiment of Grid Layout Alternation Strategy. After using Grid Layout Alternation Strategy, the shark's form of each frame remains consistent

Table II Quantitative comparison with baseline methods. We evaluate image structural similarity (SSIM), text alignment (Text Alignment), background region preservation (Masked-PSNR), and temporal consistency (Tem-Con) for each model, and analyze their characteristics through these metrics. Bold indicates the optimal choice, and underline indicates the second-best choice. ↑ denotes that higher values indicate better performance.

Method	Masked-PSNR↑				SSIM↑				Text Alignment↑				Tem-Con↑			
	8-f	16-f	48-f	200-f	8-f	16-f	48-f	200-f	8-f	16-f	48-f	200-f	8-f	16-f	48-f	200-f
Tune-A-Video [18]	13.91	15.87	15.13	—	0.4350	0.4749	0.4418	—	0.2960	0.2727	0.2894	—	0.9488	0.9481	0.9441	—
Video-P2P [20]	23.66	22.23	—	—	0.6482	0.6217	—	—	0.2937	0.2855	—	—	0.9495	0.9440	—	—
FateZero [17]	23.83	—	—	—	0.6572	—	—	—	0.2872	—	—	—	0.9576	—	—	—
RAVE [16]	22.98	20.18	20.24	20.16	0.5680	0.5349	0.5894	0.5419	0.3145	0.3128	0.3044	0.2928	0.9795	0.9753	0.9696	0.9619
ControlVideo [20]	20.60	19.97	—	—	0.5923	0.5517	—	—	0.3070	0.3012	—	—	0.9746	0.9732	—	—
Control-A-Video [17]	10.72	9.715	11.52	—	0.4802	0.3944	0.4245	—	0.2942	0.3065	0.2979	—	0.9673	0.9638	0.9506	—
TokenFlow [15]	23.55	20.71	20.03	20.38	0.6302	0.6431	0.6313	**0.6130**	0.2917	0.2980	0.2973	0.2898	0.9776	0.9712	**0.9724**	0.9718
AdaFlow [23]	23.10	21.15	20.15	19.32	0.6500	0.6499	**0.6357**	0.5819	0.2945	0.2961	0.3040	0.2708	0.9789	0.9742	0.9720	0.9517
GLFM [61]	22.91	20.49	20.45	21.19	0.6031	0.6590	0.6217	0.6017	0.3149	0.3117	0.3037	**0.2991**	0.9791	0.9677	0.9626	0.9563
UELVE-Attention	**24.05**	**22.83**	**21.78**	**21.65**	**0.6849**	**0.6635**	0.6176	0.5833	0.2841	0.2955	0.2922	0.2893	0.9653	0.9655	0.9594	**0.9725**
UELVE-ControlNet	21.11	20.05	20.60	20.02	0.5717	0.5848	0.5412	0.5578	**0.3171**	**0.3212**	**0.3058**	0.2952	**0.9803**	**0.9769**	0.9704	0.9644

Table III Time cost comparison with baseline models. UELVE serves as a plug-and-play enhancement module that can seamlessly integrate with various mainstream image editing methods, including Attention-based and ControlNet-based approaches, significantly improving their processing efficiency and long-video editing capabilities. All experiments were conducted using original videos from our dataset, UELVE-LD. For models unable to process long videos due to GPU memory constraints, we denote their results with "—".

Method	8f		16f		48f		200f	
	Inversion	Denoising	Inversion	Denoising	Inversion	Denoising	Inversion	Denoising
Tune-A-Video [18]	6s	11s	11s	22s	34s	1m08s	—	—
Video-P2P [20]	13s	48s	26s	1m38s	—	—	—	—
FateZero [17]	3m13s	42s	—	—	—	—	—	—
RAVE [16]	18s	20s	33s	37s	1m42s	1m46s	6m12s	7m38s
ControlVideo [37]	13s	17s	17s	31s	—	—	—	—
Control-A-Video [60]	7s	18s	7s	36s	7s	1m49s	—	—
TokenFlow [15]	30s	21s	59s	49s	3m14s	2m20s	12m14s	9m27s
AdaFlow [23]	7s	23s	12s	54s	37s	2m58s	2m44s	12m17s
GLFM [61]	27s	57s	51s	1m38s	2m07s	3m43s	7m51s	13m15s
UELVE-Attention	18s	18s	33s	30s	1m28s	1m21s	3m17s	5m28s
UELVE-ControlNet	6s	8s	10s	16s	26s	46s	1m45s	3m12s

synergistically optimize both the background and the edited foreground, achieving higher Text Alignment metric. GLFM [61] performs independent probabilistic clustering on each clip, achieving superior on the Text Alignment metric, but exhibits poor performance on other metrics. AdaFlow uses the pruning algorithm to handle long videos, when the frame count reaches 200, the pruning algorithm is triggered, and metric values decline.

Compared to other baseline models, our UELVE model achieves the best performance in Masked-PSNR metric across all frame settings, demonstrating its superiority in background preservation. In other metrics, our UELVE also achieves the best performance in most frame settings, showcasing robust and reliable overall performance.

Efficiency: Comparisons of time costs at different video lengths

We compare the time cost between UELVE method and baseline methods at different video lengths. As shown in Table III, compared to Attention-based editing methods, ControlNet-based methods eliminate the complex attention feature maps operations. Therefore, they exhibit lower time costs in inversion and denoising phases. FateZero [17] stores attention feature maps during the DDIM inversion stage and directly reuses the stored attention feature maps for lightweight fusion during the denoising stage. Therefore, the time cost of the inversion phase is higher than denoising phase. Control-A-Video [60] omits the DDIM Inversion operation, which inversion time recorded in table is actually the time consumption of independently editing the first frame. Tune-A-Video [18], Video-P2P [20], FateZero [17], ControlVideo [37], and Control-A-Video [60] process all frames at once, which limits their applicability to videos more than 48 frames. In contrast, UELVE and RAVE [16] process data grid by grid, thus enabling them to handle longer videos. TokenFlow [15] and AdaFlow [23] only edit the keyframes and then keyframes propagate the consistent features to all frames, reducing the GPU memory requirements

and time cost. Thus, they can handle videos more than 200 frames. GLFM [61] divide long video into short clip sequences to overcomes the GPU memory bottleneck. So that, it can handle long videos.

In the comparison with all baseline models, UELVE ranks first in processing speed across all frame settings. Although RAVE [16], TokenFlow [15], AdaFlow [23], and GLFM [61] methods can also handle long videos, they store all data in GPU memory, having a clear upper limit on processable video length. In contrast, our UELVE model employs a GPU memory swapping strategy, which allows it to theoretically handle videos of infinite length. These results demonstrate that our UELVE is an effective plug-and-play module that can enhance the processing speed and long-video handling capability of other models.

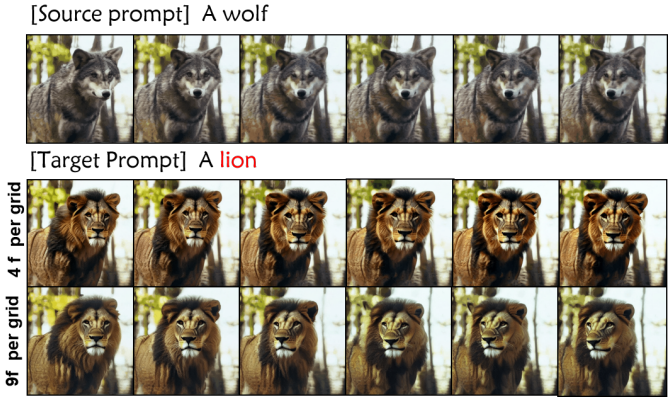

Fig. 8 The ablation experiment of grid size. With the increase of grid size, the overall beauty is improved but the details are coarse

3) User Study: To ensure the objectivity and reliability of quantitative comparisons, we also conducted a user study (detailed methodology are provided in VI-A2). As shown in Fig. 15, The results demonstrate that our UELVE model also achieved excellent performance in the user study.

C. Ablation Studies

1) Grid Layout Alternation Strategy: The serial grid batch processing method focus solely on short-term consistency within a grid, the Grid Layout Alternation Strategy is designed to address long-term consistency. For the Grid Layout Alternation Strategy ablation experiment, the source video come from our UELVE-LD dataset and is divided into 12 frames at equal intervals (grid size is 4 and grid shape is 2×2). The first two frames from each grid in the edited video are used to construct the ablation result figure. As shown in Fig. 7, without the Grid Layout Alternation Strategy, the shark in different grids presents different shapes, while with the Grid Layout Alternation Strategy, the shark between different grids present consistent shapes.

2) Grid Size: Regarding the strength of consistency maintenance, the Grid Layout Alternation Strategy is weaker than self-attention mechanism. We believe that a larger grid size will amplifies the influence of self-attention and diminish the influence of the Grid Layout Alternation Strategy. This enhances the model's grasp of the global context but coarsens the details. Fig. 8 shows that while a grid size of 9 (3×3) improves the overall aesthetic quality and lion form over a size of 4 (2×2), it also introduces obvious flaws in details. These include a black-blurred left ear and an abnormally folded right ear in some frames.

The Self-Attention mechanism by computing attention scores between every token to assess their mutual dependencies. This process results in the computational cost that increases quadratically with the image area. For instance, processing a frame of size $H \times W$ (in the latent space) incurs the computational cost of approximately $O(H^2W^2)$. If we process all N frames at once, the computational cost is approximately $N^2 \cdot O(H^2W^2)$. If we first transform the sequence of N frames into the sequence of N/n grids (each grid containing n frames) and then serially process each grid (the Grid Layout Alternation Strategy can maintain the consistency cross grids), the computational cost is approximately $N \cdot n \cdot O(H^2W^2)$, which is significantly less than processing all N frames at once. From this perspective, fewer frames per grid results in higher computational efficiency.

However, when the grid size is smaller (i.e., having less frames per grid and having more grids), the frequency of I/O operations and model parameter loading increase, the girds communication synchronization expands, and the fragmentation issues becomes more severe, which lead to a linear growth in cost time. As shown in Fig. 9, the time consumed does not linearly increase or decrease with the number of frames per grid. Instead, there is a local optimum. The local optimum is achieved when there are 6 frames per grid. Since the fine-tuning phase executes 300 time steps while the inversion and denoising phases only execute 50 time steps, the tune phase get the greater fluctuations in time consumption.

3) fine-tuning: After integrating the temporal attention and fine-tuning method, the model successfully transitions from a static image paradigm to a dynamic spital-temporal coherent video paradigm. The static-dynamic association allows static details to adapt to the dynamic video scene and makes frame content editing far more flexible, no longer forced to rigidly adhere to the contour details of the original image.

As shown in Fig. 12, before fine-tuning, although the model accomplishes local editing from a rabbit to a cat, the contour structure of rabbit persist. After fine-tuning, the model not only executes the local edit from a rabbit to a cat, but also achieves a complete transformation of the contour structure from a rabbit to a cat.

4) Necessity of reverting the data form: The parameters in stable diffusion model are trained based on the frame image, but at this time we want to use it to predict the noise of the grid image. Therefore, as shown in Fig. 11, we consider modifying the logic of the Basic Transformer Block in diffusion model to adapt the gird image.

Firstly, we combine the frame sequence into a grid in fine-tuning stage to be consistent with the denoising process. As shown in Fig. 10, With the grid training, the overall picture is clearer and the tone is more natural, especially in the light of the woods in the background.

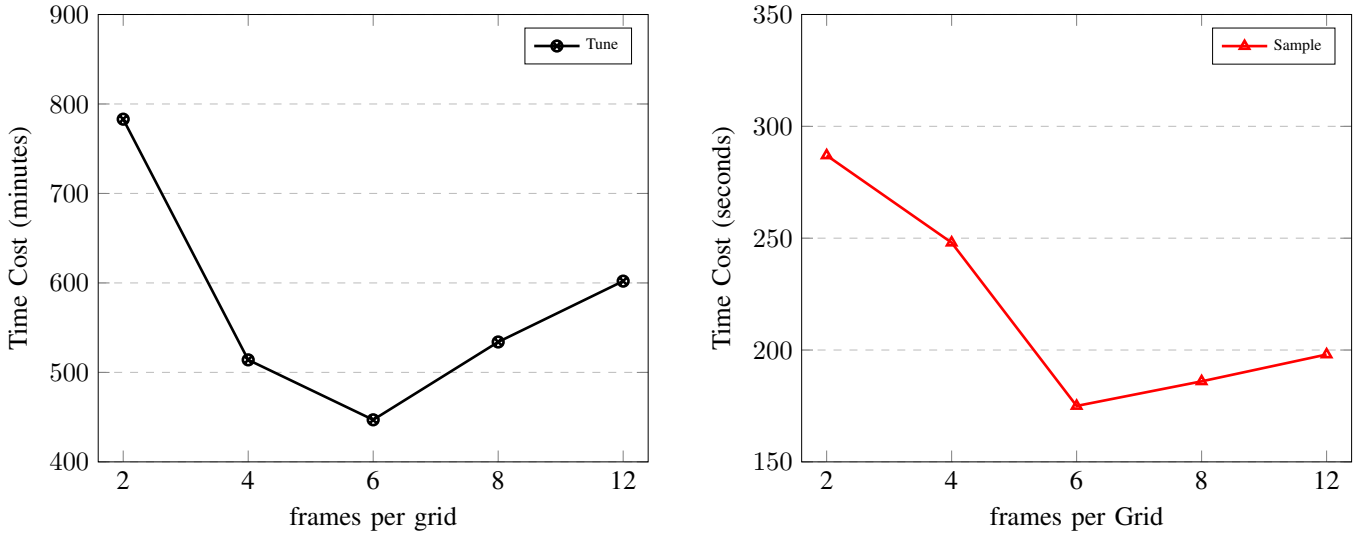

Fig. 9 Time cost analysis for different grid size. The video material consists of 48 frames and comes form our UELVE-LD dataset. The left picture is Tuning time cost across different grid sizes; The right picture is Sample times across different grid sizes. The sample stage include inversion and denoising stage. The grid shape (H $\times$ W) corresponding to different grid size are as follows: 2 frames = 2×1, 4 frames = 2×2, 6 frames = 2×3, 8 frames = 2×4, and 12 frames = 3×4.

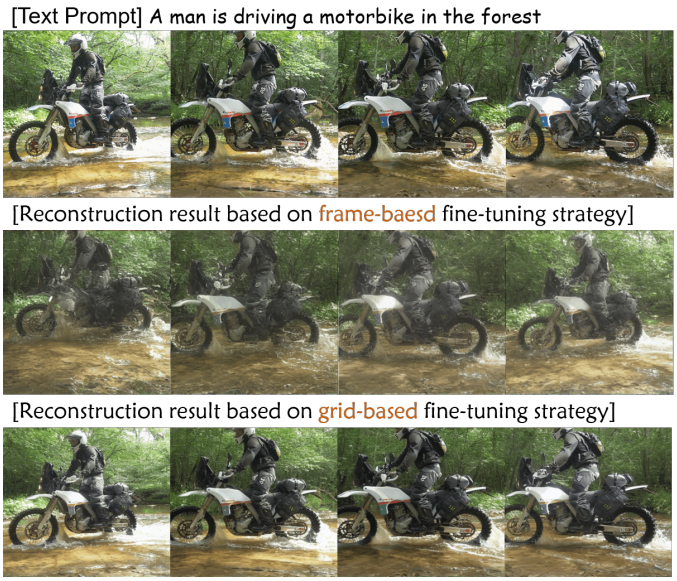

Fig. 10 Ablation Grid tune. Grid tune lets the model parameters used in tune stage and sample stage consistent to achieve better editing effect.

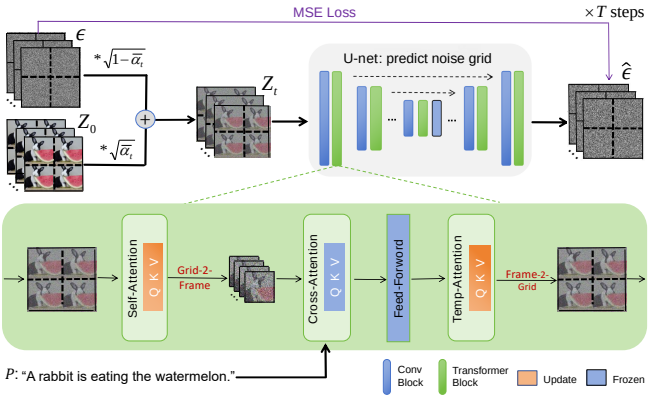

Fig. 11 To adapt to the grid elements, we have modified the processing logic of the BasicTransformerBlock in the original U-Net noise prediction model. The key modifications are threefold. 1.Replaced the original single-frame input with a grid-based input format. 2.Introduced a Grid-to-Frame (G2F) operation after the self-attention layer and a Frame-to-Grid (F2G) operation after the temporal attention layer.

Then, since mask construction requires interpolating cross attention maps to latent variable dimensions, numerical values at edge regions of attention maps from different frames may interfere with each other during interpolation, leading to degraded mask quality. After completing the self-attention, we split the grid feature maps into frame sequences and reordered them for cross-attention input, ensuring attention maps contain only single-frame information to avoid interference during mask construction. As shown in Fig. 13, with the frame image based cross attention input, penguin masks demonstrates significant quality improvement after splitting frame grid to frames, with previously affected areas fully recovered.

V. Conclusion

In this paper, we present UELVE, an LLM-driven unified framework with parallel grid diffusion for efficient long video editing. Our approach integrates with an LLM, through the semantic analysis and reasoning to combine the advantage of various video editing type and achieve the basis of unified video editing framework. Specially, we utilize grid self attention to enhance the short-term consistency among intra-

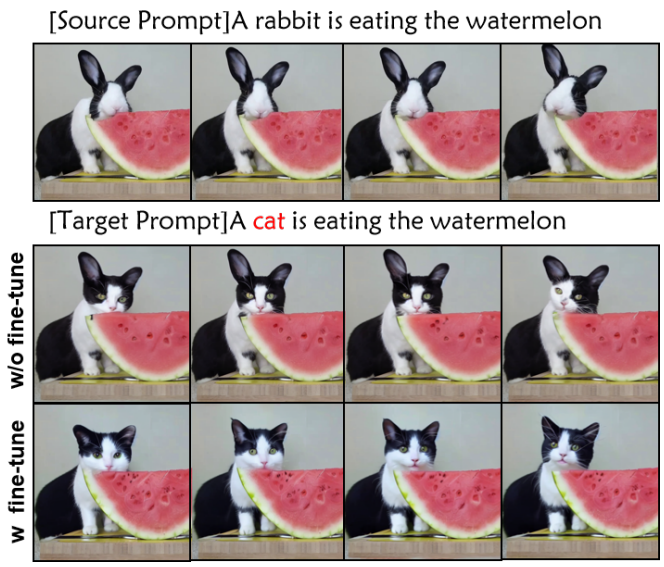

Fig. 12 The ablation of fine-tuning operation. Fine-tuning enhances the flexibility of editing in local editing tasks, making the results more faithful to the semantic meaning of the text prompt.

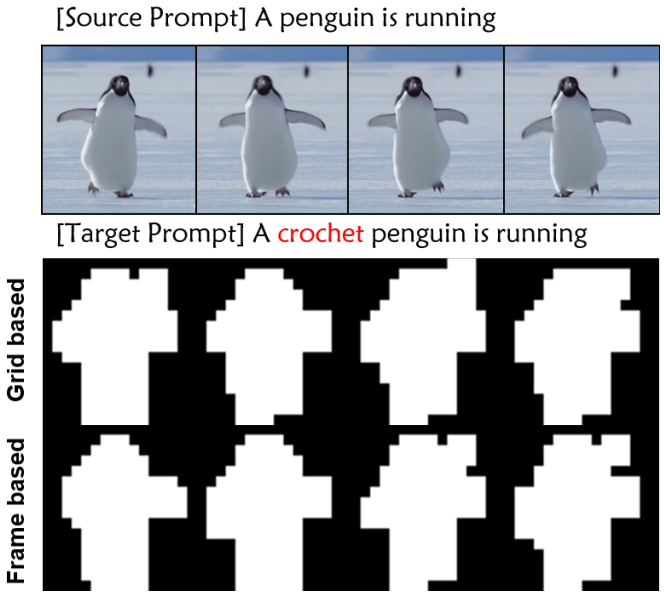

Fig. 13 The ablation of Grid mask. In the processing of generating masks, the grid sequences are converted into frame sequences to avoid the mutual influence between frames in the grid.

grid neighboring frames and design a grid layout alternation strategy to maintain long-term consistency of the whole video. we propose the grid batch diffusion operation and multi-GPU data parallel computing method to significantly improve computational efficiency. In addition, our UELVE can be used as a plugin, combined with various existing image editing techniques (such as attention control [38] or ControlNet[39] based), to achieve desirable video editing. Although the temporal awareness module enables the handling of simple motion editing tasks, effectively completing complex cases remains a challenge. In future work, we plan to integrate specialized motion editing methods to overcome this limitation.

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VI. Appendix

A. Appendix

1) Evaluation Metrics For Details: The following evaluation metrics were utilized to assess the quality of edited videos.

SSIM (Structural Similarity Index Measure) Meaning: This metric quantifies the structural similarity between two images. Calculation: SSIM is used to measure the similarity by comparing three key attributes (Luminance, Contrast and Structure) between the frame image of source video and edited video. It aligns more closely with human visual perception than traditional metrics like MSE and PSNR. Optimal Value: Ranges between [0, 1], with higher values indicating better similarity.

Masked PSNR (Masked Peak Signal-to-Noise Ratio) Meaning: This metric is used to assess the background preservation of method Calculation: It computes the Peak Signal-to-Noise Ratio (PSNR) in designated mask regions (most often the background) between the source video and its edited counterpart. Optimal Value: Higher values are preferable.

Text Alignment Meaning: This metric evaluates the similarity between generated videos and their corresponding textual descriptions. Calculation: It computes the cosine similarity between video frames and their text description using the CLIP model. Optimal Value: Higher values indicate better alignment.

Temporal Consistency Meaning: This metric assesses the continuity of features between adjacent video frames. Calculation: It computes the cosine similarity of features between neighboring frames. The value of this metric monotonically increase with the smoothness about the edited object's motion. Optimal Value: Higher values denote greater consistency.

These metrics collectively provide a comprehensive evaluation framework, addressing background preservation, structural fidelity, text alignment, and temporal coherence in edited visual content.

2) User Study: Because the quantitative metrics values cannot replace human visual perception, we conducted a user study to get a more fair and objective comparison. To ensure the comprehensiveness and fairness of the study, we designed a structured framework consisting of four key questions. Participants were asked to select the best edited video for each question.

The four questions were as follows: Overall editing effect: Which specific edits would you consider to be the most successful in general?

Temporal consistency: please select which have the highest level of maintaining the motion of source video.

Text alignment: Which video most closely corresponds to the text prompts provided above?

Background preservation: please choose the editing videos that preserve the original background best.

As shown in the pie charts in Fig. 15, our model performed excellently across all four survey questions, receiving recognition and positive feedback from the vast majority of users.

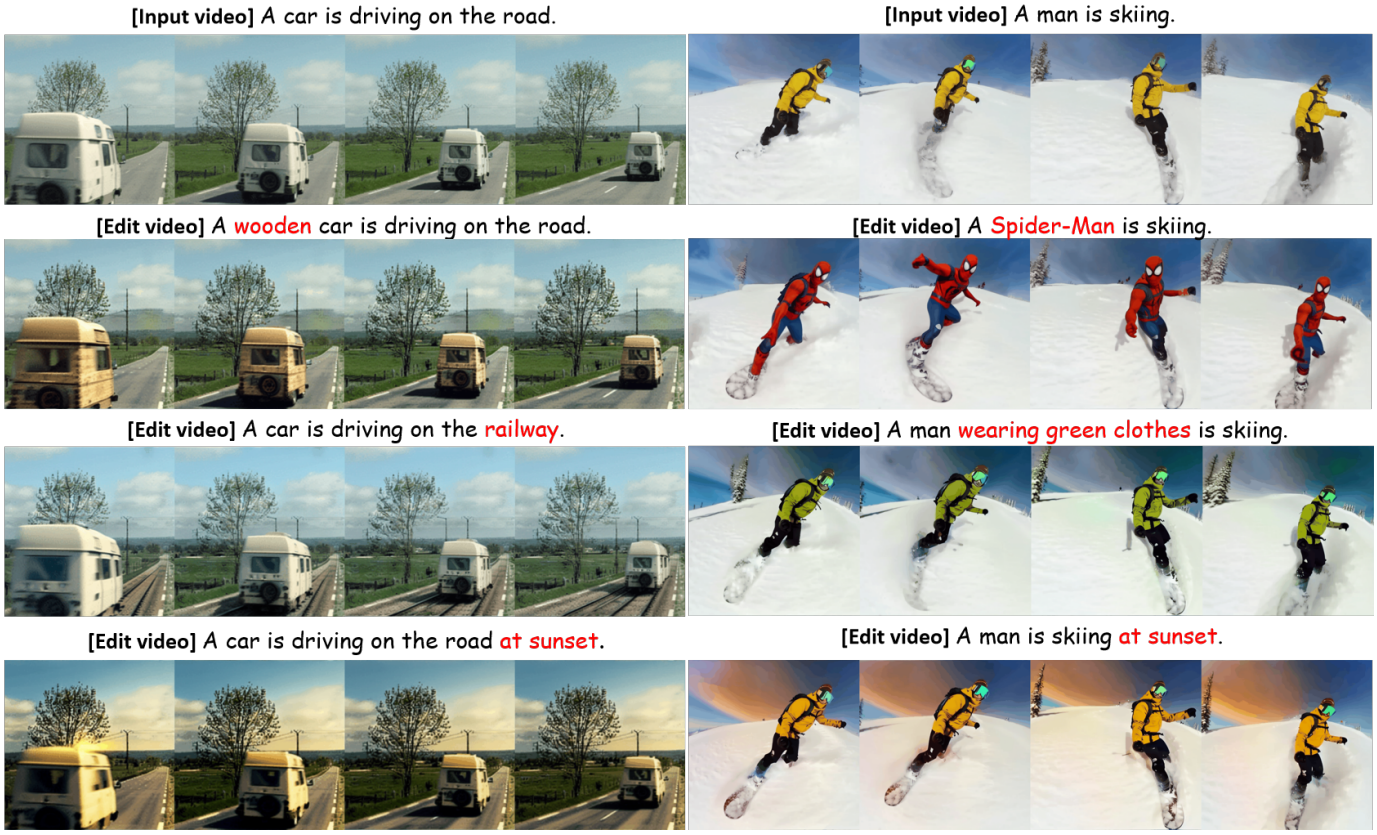

Fig. 14 The editing example directed by DeepSeeK.

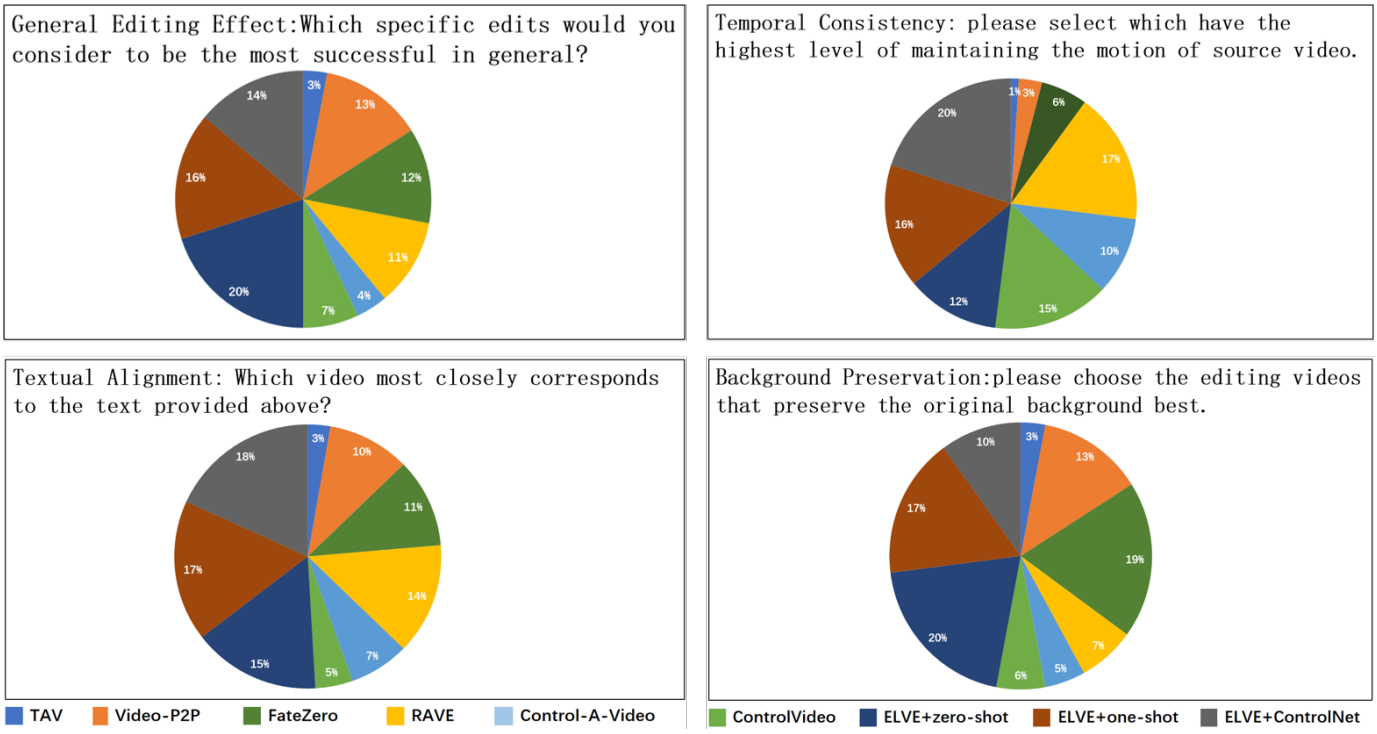

Fig. 15 The pipe charts of user-study

3) Transfer Volume Between GPU: This section provides a clear comparison of the interaction demands between the Alternate-based multi-GPU data parallelism method and the Shuffle-based multi-GPU data parallelism method. We employ both theoretical analysis and experimental validation to illustrate the differences in data transfer volume.

Theoretical Analysis: For data-parallel processing of a video containing N frames across K ($K > 1$) GPUs, both the Alternate-based method and the Shuffle-based method configure the grid shape as $H \times W$, satisfying $H \times W \leq N/K$ and $N\%(H \times W) = 0$. Initially, the grid sequence is uniformly distributed across all GPUs, with each GPU storing $N/(K \times H \times W)$ grids.

For Shuffle-based method, the system first generates a random permutation sequence $\pi = [p_1, p_2, \ldots, p_N]$ of all N frames, then redistributes the frames according to this new sequence. Specifically, GPU j (where $j = 0, 1, \ldots, K-1$) ultimately receives the frame set:

$$\mathcal{F}_j = \{p_{jM+1}, p_{jM+2}, \ldots, p_{(j+1)M}\}$$

where $M = N/K$.

The time cost of the redistribution process is dominated by inter-GPU data transfer. Frames whose final location differs from their initial location must be transmitted across the interconnect network. Our primary objective is to quantify the expected volume of this data transfer.

Let X denote the total number of frames transferred. We analyze its expected value using probabilistic methods. Define an indicator random variable for each frame i:

$$I_i = \begin{cases} 1 & \text{if frame } i \text{ is transferred (initial GPU} \neq \text{final GPU)} \\ 0 & \text{otherwise} \end{cases}$$

The total transfer volume can be expressed as $X = \sum_{i=1}^{N} I_i$. By linearity of expectation:

$$\mathbb{E}[X] = \sum_{i=1}^{N} \mathbb{E}[I_i] = \sum_{i=1}^{N} \Pr(I_i = 1)$$

Under the uniform random permutation, each frame has an equal probability of landing in any specific GPU in the final allocation. Since each frame initially resides in exactly one specific GPU, the probability that it remains in the original GPU after redistribution is precisely $1/K$. Therefore, the probability that a frame needs to be transferred is:

$$\Pr(I_i = 1) = 1 - \frac{1}{K}$$

Substituting into the expectation formula yields the main result:

$$\mathbb{E}[X] = \sum_{i=1}^{N} \left(1 - \frac{1}{K}\right) = N\left(1 - \frac{1}{K}\right)$$

So, the Shuffle-based methods use K GPUs to handle N frames video with $H \times W$ grid shape, the total data exchange volume is $N\left(1 - \frac{1}{K}\right)$ frames.

For Alternate-based Method, the GPUs only need to exchange H frames at the breakpoints of the grid sequence. For K GPUs, the total data exchange volume is (K-1)H frames. As shown in Fig 4, it is the case of two GPUs.

Experiments validation: The foundational hardware for this experiment is a server equipped with two GPUs, interconnected via a SYS topology (connection traversing PCIe as well as the SMP interconnect between NUMA nodes). On the software side, we use the NVIDIA Collective Communications

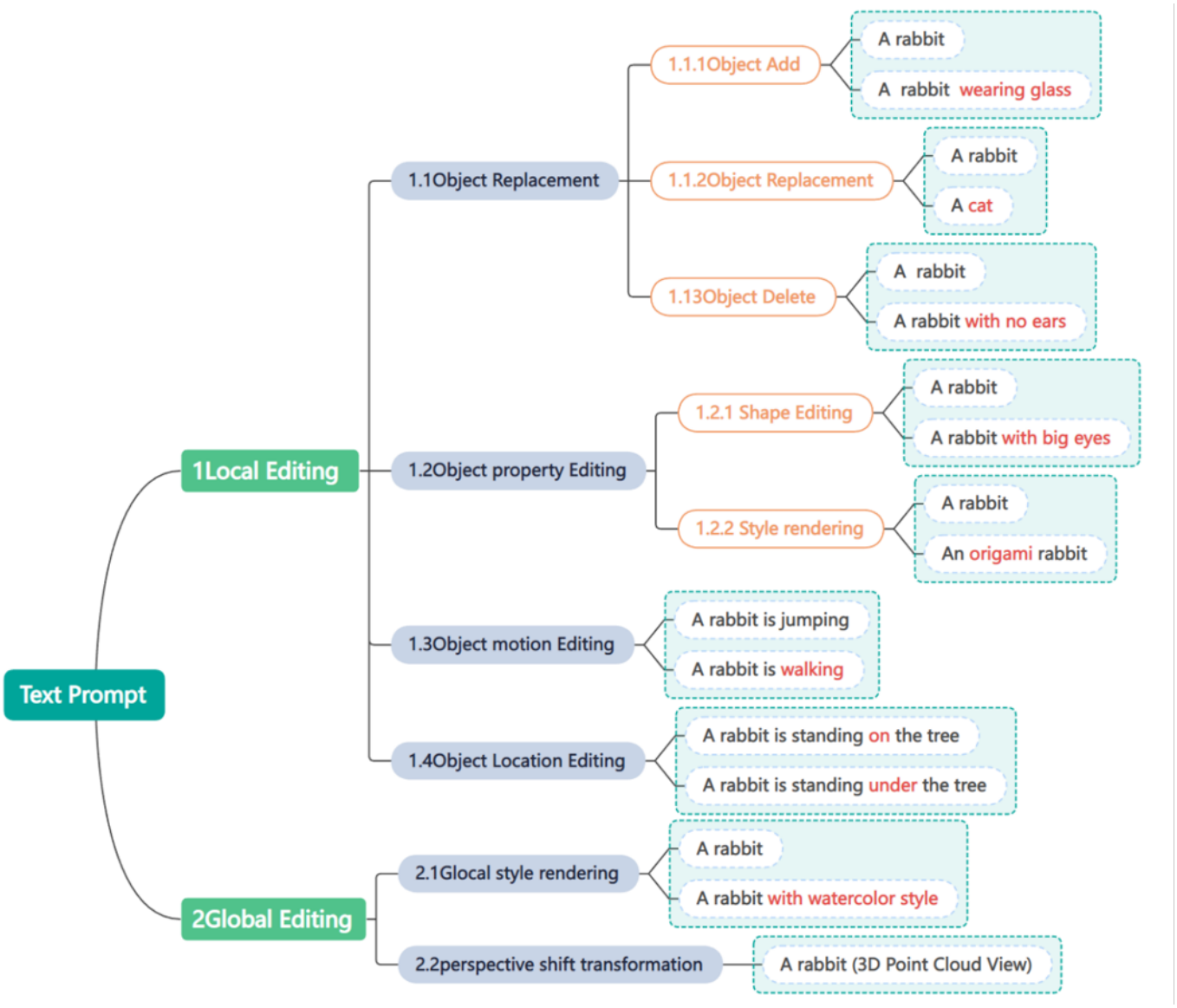

Fig. 16 The classification standard for Text Prompt.

Library (NCCL) for high-performance inter-GPU communication and the Hugging Face Accelerate library serves as the data-parallel framework for distributing the workload across the GPUs. For the experimental evaluation, we selected a subset of 12 videos from the UELVE-LD dataset, comprising three 16-frame videos, three 32-frame videos, three 64-frame videos, and three 96-frame videos. The video editing tasks were completed using either the Alternate-based or Shuffle-based multi-GPU data-parallel method, with a common grid shape of 2×2 for both approaches. We recorded the data transfer volume between GPUs for each experiment and calculated the average value across all video samples as the final result. As shown in Table IV, the data transfer volume for the Alternate-based method remains fixed at 2 frames, while that for the Shuffle-based method fluctuates around half the number of frames in each video. In summary, the experimental results validate our theoretical analysis.

Both the mathematical theoretical analysis from the perspective of probability theory and the experimental validation demonstrate that the inter-GPU data transfer requirement is $N\left(1-\frac{1}{K}\right)$ frames for the Shuffle-based method, compared to (K-1)H frames for the Alternate-based method.

Table IV The communication demands between GPUs with Alternate-based method and Shuffle-based method.

Method	16f	32f	64f	96f
Alternate-Based	2	2	2	2
Shuffle-Based	8	16	31	47

4) The Layout Selection: In addition to Layout 1 introduced in Section III-D, As shown in Fig. 5 . we have proposed two other layouts: Layout 2, Layout 3, As shown in the Fig. 18, all set the grid shape as 2×2.

Layout 2 is similar to Layout 1 and it has a simpler structure. However, it introduces two new issues: on one hand, it doubles the communication requirements between GPUs; On the other

Now please help me analyze the following: source prompt: A man is skiing on the mountain , target prompt: A Spider-Man is skiing on the mountain. Which of these text prompts best describes the editing behavior and fixed categories? Is it necessary to use a mask mechanism? Should fine-tuning be applied during training? Generate a new configuration file and analyze the setting and rationale of key parameter values

Editing Behavior Analysis

Source Prompt: A man is skiing on the mountain
Target Prompt: A Spider-Man is skiing on the mountain

1. Fixed Category:

- **1.1.2 Object Replacement**
  - The edit replaces the "man" with "Spider-Man", altering the main subject (man).

2. Mask Mechanism:

- ✅ **Required** (Local edits need mask)

3. Fine-Tuning (Tune):

- ✅ **Required** (Structure been changed)

```python
# configuration file
use_invertion_latents: true
use_inversion_attention: true
grid_sizeH: 2  grid_sizeW: 2
is_replace_controller: True
cross_replace_steps: default_: 0.2
self_replace_steps: 0.3
eq_params:
          words: ["Spider-Man"]
            values: [3]
blend_words: [['man'],['Spider-Man']]
blend_self_attention: True    blend_self_th: [0.4, 0.4]
blend_latents: True  blend_latents_th: [0.5,0.5]
```

Fig. 17 The semantic analysis of text prompt by DeepSeeK.

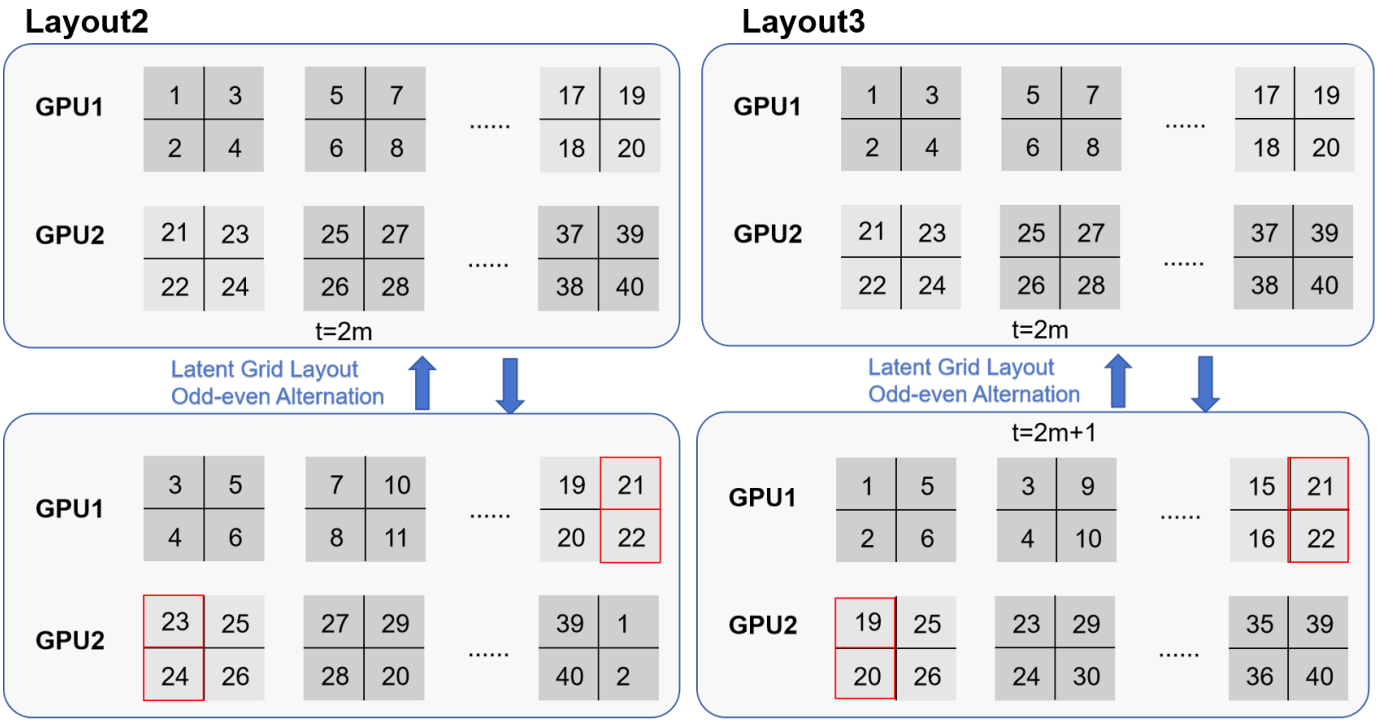

Fig. 18 The structured figure of Layout 2 (Left) and Layout 3 (Right)

hand, it places the first two and the last two frames within the same grid, which results in inferior performance when editing short videos. Layout 3 directly swaps adjacent frames between grids, which not only double the inter-GPU communication requirements, but also perform worse in short video.

Overall, Layout 1 is suitable for both short and long videos. In contrast, Layout 2 and Layout 3 are only appropriate for long videos. Empirical results show that Layout 2 performs well with sequences of 96 frames or more, while Layout 3 performs well with sequences of 48 frames or more. As shown in Fig. 19, when the video is short (Left), Layout 1 performs well, Layout 2 performs poorly at the beginning and end but relatively well in the middle, and Layout 3 perform poorly overall. When the video is long (Right), all of three Layouts perform well.

5) DeepSeek For UELVE: we first use LLM (e.g., DeepSeek, implemented via API interfaces) to semantically analyze the text prompt and then classify the text prompts based on the Editing Type Classification standard we provide (as shown in Fig. 16), and finally make the correct decision. In addition, our UELVE can intelligently recommend values for parameters by LLM, As shown in Fig. 17.

To more intuitively demonstrate how DeepSeek performs

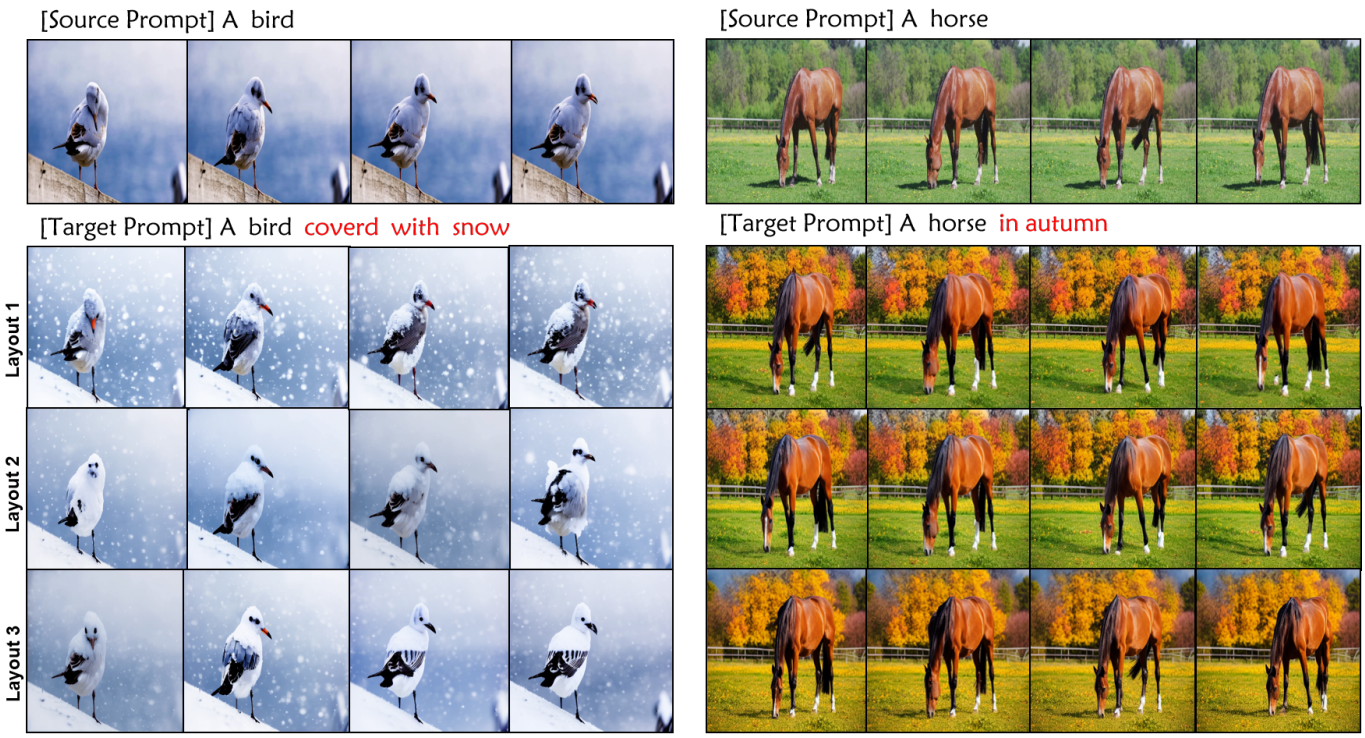

Fig. 19 The editing result of different Layout, Left is short video, right is long video (Layout 2 rubbish the first and end section of edited video).

text prompt analysis, Text Prompt classification, decision making and parameter recommendation tasks based on predefined criteria, we directly use DeepSeek by the official DeepSeek website. As shown in Fig. 17, the text prompt classification, decision-making, and parameter recommendations align with our expectations. Figure 14 shows a great editing result obtained using the parameters recommended by DeepSeek.

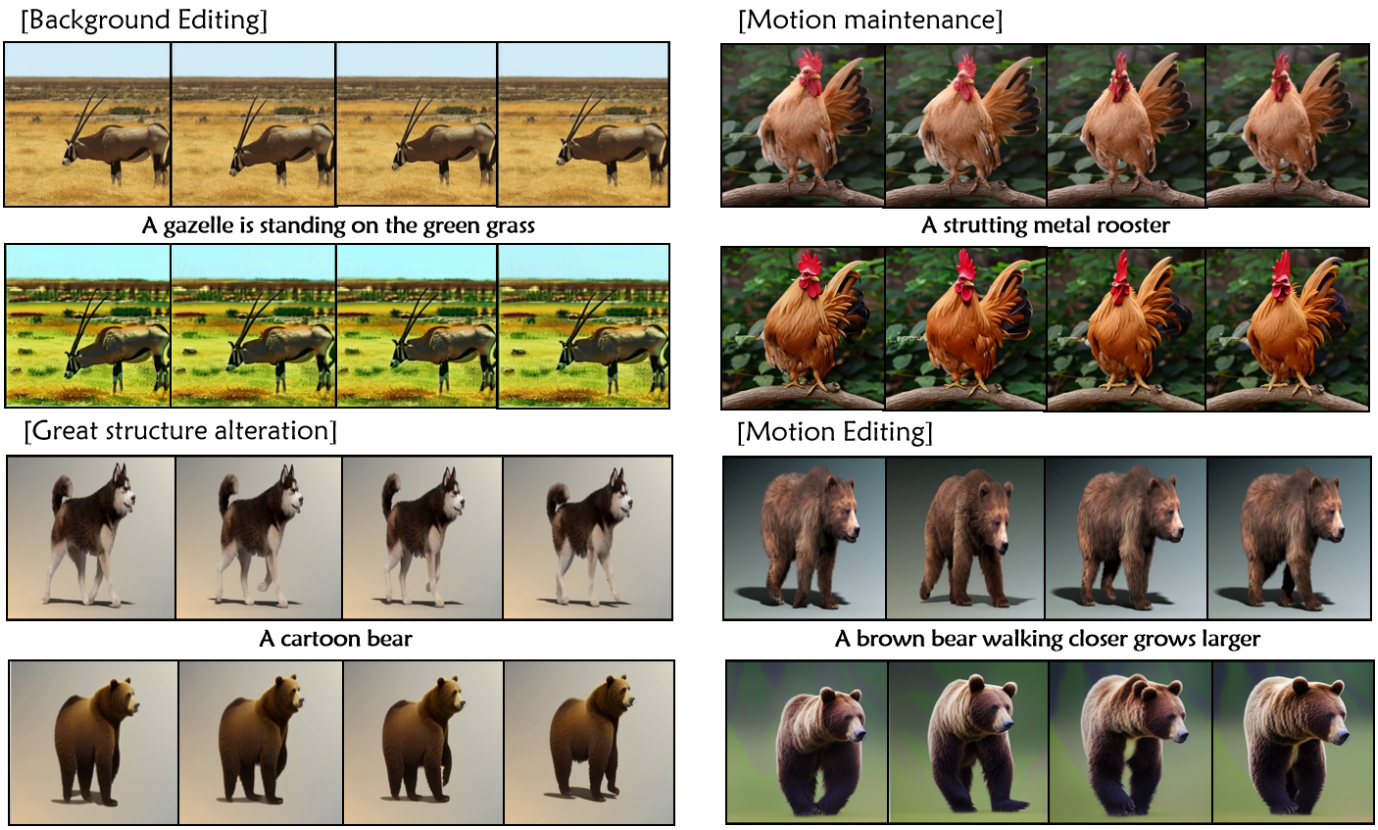

Fig. 20 UELVE is a unified video editing framework capable of handling various specialized editing tasks that are normally achieved by models with specific architectures.